

The mean

The arithmetic mean, commonly called the average, is the sum of the values divided by their number.

$$\bar{x} = \frac{\sum x}{n}$$

What is the mean of synthetic counts 2, 3, 5, 6?

A mean with units

Synthetic file sizes: 10, 15, 20, 35 KB.

What is their mean, including units?

Recover a total

Five synthetic downloads have a mean duration of 8 seconds.

What is their total duration?

Sample and population means

$\bar{x}$ denotes a sample mean; μ denotes a population mean.

Which symbol describes the mean size of all files on a server?

The median

Synthetic data. The median is the middle ordered value, or the average of the two middle values.

What is the median of 9, 2, 7, 4, 3?

An even-sized sample

Synthetic downloads per session: 0, 1, 2, 2, 4, 9.

What is the median?

Median position and value

The median position is $(n + 1)/2$ in ordered data. M can denote the median value. A synthetic sample contains 40 values.

Which positions determine M , and why is 20.5 not necessarily its value?

The mode

Synthetic data. A mode is a value with greatest frequency.

What is the mode of 0, 1, 2, 2, 2, 4, 4?

Two modes

A bimodal dataset has two values tied for highest frequency. Synthetic counts: 0, 1, 1, 3, 3, 4.

Which values are modes?

A categorical mode

Synthetic notebook colors: blue, black, blue, red, black, blue.

Which measure of center is meaningful for these categories?

Compare mean and median

Synthetic file sizes (KB): 4, 5, 5, 6, 30.

How do the mean and median compare?

Sensitivity to one value

Synthetic file sizes (KB): 4, 5, 5, 6, 30. Replace 30 with 300.

Which measure of center changes more?

Choosing a typical value

Most files are small, but a few archive files are enormous.

Which center would better describe a typical file?

A mean from frequencies

Synthetic print-job counts.

Pages	Jobs
1	2
2	3
5	1

What is the mean number of pages per job?

Why weight the values?

Synthetic job frequencies: one page 2 times, two pages 3 times, five pages 1 time.

Why is $(1 + 2 + 5)/3$ not the mean per job?

A grouped estimate

A frequency table records frequencies for values or intervals. An interval midpoint is the average of its boundaries. Synthetic file sizes:

Size (KB)	Frequency
0 to <10	3
10 to <20	5
20 to <30	2

What midpoints would you use to estimate the mean?

Estimate a grouped mean

Synthetic file sizes: intervals $[0,10)$, $[10,20)$, $[20,30)$ have frequencies 3, 5, 2.

What mean results from weighting the class midpoints?

Exact or estimated?

A grouped table gives only interval counts, not individual values.

Why is a midpoint-based mean generally an estimate?

Combine two groups

Synthetic group A: 10 jobs, mean 4 pages.
Group B: 30 jobs, mean 8 pages.

What is the mean for all 40 jobs?

Adding a constant

Every synthetic print job gains a two-page cover sheet.

How does the mean pages per job change?

Changing units

Synthetic file sizes have mean 2 MB. Use 1 MB = 1,000 KB for this problem.

What is the mean in KB?

Law of Large Numbers

For independent observations from the same population with a finite mean, larger samples make the sample mean more likely to be near the population mean.

Does this guarantee that every added observation improves the estimate?

Sampling distribution

A sampling distribution describes a statistic over repeated random samples of the same size.

If we compute one mean per sample, what does each point in its sampling distribution represent?

A statistic estimates a parameter

A sample statistic, such as $\bar{x}$, estimates a population parameter, such as μ .

Why would the distribution of sample means differ from the distribution of individual values?